

1. Protocol Synopsis & Overview

Protocol Title: A Phase II, Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Compound X in Patients with Moderate-to-Severe Plaque Psoriasis.

Study Drug: Compound X (100 mg) **Phase:** II **Indication:** Plaque Psoriasis

2. Materials and Methods

2.1 Study Design

This was a multicenter, randomized, double-blind, placebo-controlled, parallel-group study conducted to evaluate the efficacy, safety, and tolerability of Compound X compared to placebo over a 12-week treatment period.

Patients who met the eligibility criteria were randomized in a 1:1 ratio to receive either:

- Group A:** Compound X 100 mg administered orally once daily (QD).
- Group B:** Matching placebo administered orally once daily (QD).

2.2 Efficacy Endpoints

The primary efficacy endpoint was the proportion of patients achieving a $\geq 75\%$ reduction from baseline in the Psoriasis Area and Severity Index score (PASI 75) at Week 12.

The secondary endpoints included:

- The proportion of patients achieving a Static Physician's Global Assessment (sPGA) score of "clear" (0) or "almost clear" (1) at Week 12.
- Change from baseline in the Dermatology Life Quality Index (DLQI) at Week 12.

2.3 Statistical Analysis

Efficacy analyses were performed on the Intent-to-Treat (ITT) population, defined as all randomized patients who received at least one dose of the study drug. Categorical variables were analyzed using the Chi-square test, and continuous variables were analyzed using an Analysis of Covariance (ANCOVA) model with baseline values as covariates.

3. Study Population & Demographics

A total of 120 patients were randomized (60 to the Compound X group, 60 to the Placebo group). Of these, 115 patients completed the 12-week treatment period. Five patients discontinued prematurely (2 due to adverse events in the Compound X group, 3 due to lack of efficacy in the Placebo group).

Demographic Characteristic	Compound X (N=60)	Placebo (N=60)
Age (Years), Mean (SD)	44.5 (11.2)	43.1 (12.4)
Male, n (%)	34 (56.7%)	32 (53.3%)

Baseline PASI Score , Mean (SD)	18.4 (4.1)	17.9 (3.8)
--	------------	------------

4. Results

4.1 Primary Efficacy Outcomes

At Week 12, a significantly higher proportion of patients in the Compound X group achieved PASI 75 compared to the placebo group.

- **Compound X:** 73.3% (44/60 patients) achieved PASI 75.
- **Placebo:** 11.7% (7/60 patients) achieved PASI 75.

The treatment difference was statistically significant with a p-value of $p < 0.001$.

PASI 75 Response Rates at Week 12:



4.2 Secondary Efficacy Outcomes

The secondary endpoint of sPGA score of 0 or 1 at Week 12 supported the primary findings. Clear or almost clear skin was achieved by 65.0% of patients in the Compound X arm compared to 8.3% in the placebo arm ($p < 0.001$).

5. Safety Evaluation

Compound X was generally well-tolerated. The overall incidence of treatment-emergent adverse events (TEAEs) was similar between the treatment groups (45.0% for Compound X vs. 41.7% for placebo).

- **Most Common TEAEs ($\geq 5\%$):** Mild headache (8.3%) and nausea (6.7%) in the Compound X group.
- **Serious Adverse Events (SAEs):** No treatment-related SAEs or deaths were reported during the 12-week study period.

6. Discussion and Conclusions

The data from this Phase II study demonstrate that Compound X (100 mg QD) provides statistically significant and clinically meaningful improvements in skin clearance compared to placebo in patients with moderate-to-severe plaque psoriasis. The safety profile suggests an acceptable tolerability margin, supporting progression to Phase III confirmatory trials.